

EVEN-PLUS Syndrome: A Comprehensive Disease Characterization Report

Disease: EVEN-PLUS Syndrome **OMIM:** #616854 | **Gene:** HSPA9 (mortalin/mtHSP70/GRP75) | **Category:** Mendelian (autosomal recessive) **Suggested MONDO:** MONDO:0014784

Evidence types are marked where useful: human clinical, model organism, in vitro/biophysical, computational. No primary datasets were provided; all content is derived from primary literature via PubMed.

Summary

EVEN-PLUS syndrome is an ultra-rare, autosomal-recessive, prenatal-onset multiple-malformation and skeletal-dysplasia syndrome caused by biallelic loss-of-function or hypomorphic variants in *HSPA9*, the gene encoding the mitochondrial HSP70 chaperone mortalin (mtHSP70/GRP75). The name is an acronym for its cardinal features — Epiphyses, Vertebrae, Ears, Nose — **PLUS** associated malformations of the heart, kidneys, and central nervous system. The disorder was delineated and named by Royer-Bertrand and colleagues in 2015 ([PMID: 26598328](#)), who identified biallelic *HSPA9* mutations in affected individuals lacking mutations in *LONP1* (the gene responsible for the phenotypically overlapping CODAS syndrome). Together with CODAS, EVEN-PLUS defines a family of "mitochondrial chaperonopathies."

Mechanistically, mortalin is an essential mitochondrial matrix chaperone that drives ATP-dependent import of nuclear-encoded proteins across the inner membrane, assists protein folding, participates in iron-sulfur (Fe-S) cluster biogenesis, and buffers oxidative stress. EVEN-PLUS mutations — spanning missense, nonsense, frameshift, and splice-site classes — reduce or abolish this chaperone activity. Biophysical work on the nucleotide-binding-domain mutants R126W and Y128C shows that they disrupt ATP hydrolysis, interdomain communication, and thermostability while increasing the protein's propensity to aggregate ([PMID: 30933555](#)). The downstream consequence — impaired mitochondrial proteostasis, oxidative stress, and apoptosis in rapidly dividing embryonic precursors — is inferred to explain the skeletal,

craniofacial, and organ malformations. Complete loss of the orthologous gene is embryonic-lethal in mice and produces ineffective hematopoiesis in zebrafish, establishing mortalin as developmentally essential.

Clinically, fewer than ~15 patients have been reported worldwide. Onset is prenatal; survivors have lifelong static skeletal dysplasia and craniofacial anomalies, with a subset developing seizures, developmental delay, and basal-ganglia lesions. Diagnosis rests on exome/genome sequencing (identifying biallelic *HSPA9* variants) combined with characteristic skeletal imaging. There is **no disease-modifying or curative therapy**; management is supportive and multidisciplinary, and prevention is reproductive — prenatal diagnosis and preimplantation genetic testing (PGT), the latter demonstrated to successfully block transmission and yield a healthy birth (PMID: 38281662).

1. Disease Information

EVEN-PLUS syndrome is a congenital, autosomal-recessive multisystem malformation syndrome with skeletal dysplasia. It was first delineated as a distinct entity in 2015, when biallelic *HSPA9* mutations were identified in three individuals from two families (PMID: 26598328). The name encodes the four cardinal anatomical domains — Epiphyses, Vertebrae, Ears, Nose — plus associated ("PLUS") malformations. The original report states the phenotype *"included severe microtia, nasal hypoplasia, and other malformations, and for which we propose the name of EVEN-PLUS syndrome for epiphyseal, vertebral, ear, nose, plus associated findings"* (PMID: 26598328).

Key identifiers:

Resource	Identifier
OMIM (disease)	#616854
OMIM (gene)	*600548 (HSPA9)
Suggested MONDO	MONDO:0014784
HGNC (gene)	HGNC:5244
Gene locus	5q31.2

Synonyms / alternative names: EVEN-PLUS syndrome; Epiphyseal, vertebral, ear, nose, plus associated findings syndrome; EVE dysplasia (the original family reported as "EVE dysplasia" was later confirmed to carry a homozygous *HSPA9* variant, [PMID: 35779070](#)).

Data provenance: All information is derived from aggregated, disease-level resources and individual published case reports/case series ($n < 15$ patients worldwide), not from EHR-derived population cohorts. This is characteristic of an ultra-rare Mendelian disorder.

2. Etiology

Primary cause — genetic: EVEN-PLUS is a monogenic disorder caused by **biallelic (homozygous or compound heterozygous) pathogenic variants in *HSPA9***, which encodes mortalin (mtHSP70/GRP75), a mitochondrial chaperone. The seminal study reported: "*we found biallelic mutations in HSPA9, the gene that codes for mHSP70/mortalin, another highly conserved mitochondrial chaperone protein essential in mitochondrial protein import, folding, and degradation*" ([PMID: 26598328](#)).

Genetic risk factors: The disorder is fully penetrant given biallelic pathogenic genotypes; there are no known susceptibility loci or modifier genes described. Consanguinity and founder alleles increase risk in specific families/populations — the frameshift variant c.882_883delAG "*may have a higher distribution frequency in East Asian populations*" ([PMID: 36052765](#)).

Environmental risk factors: None identified. This is a purely Mendelian disorder; there is no evidence of environmental, infectious, or lifestyle contribution to causation.

Protective factors: No genetic or environmental protective factors are described. Because complete mortalin loss is embryonic-lethal (mouse homozygous knockout, [PMID: 25550197](#)), viable EVEN-PLUS genotypes are inferred to be hypomorphic — retaining partial chaperone function — which acts as an implicit constraint on the survivable disease spectrum.

Gene–environment interactions: None reported or expected for this monogenic disorder.

3. Phenotypes

The phenotype spectrum is best defined by the 12-case review of Liu et al. (PMID: 38284453; 9 females) and the first-affected-male report (PMID: 32869452). Onset is prenatal/congenital; the phenotype is largely static (structural malformations) except for progressive neurologic features in a subset.

Near-universal features (~100%)

"All patients had synophrys or arched eyebrows, hypoplastic or dysplastic ears, hypoplastic nasal bone, and dysplastic femoral head." (PMID: 38284453)

Phenotype	HPO term	Type
Synophrys	HP:0000664	Physical/craniofacial
Arched eyebrows	HP:0011003	Physical/craniofacial
Microtia / hypoplastic-dysplastic ears	HP:0008551, HP:0000369	Physical/craniofacial
Hypoplastic nasal bone	HP:0004646	Skeletal/radiographic
Dysplastic femoral head / epiphyseal dysplasia	HP:0006411, HP:0002656	Skeletal

Common features (frequent)

Phenotype	HPO term
Triangular nares	HP:0011832
Bifid/dysplastic femur ("fork-shaped" distal femur)	—
Dysplastic epiphyses at the knee	HP:0002656
Dysplastic acetabula	HP:0003182
Delayed ossification	HP:0002662
Short stature	HP:0004322
Vertebral (coronal) clefting	HP:0003417
Scoliosis	HP:0002650
Dislocated patellae	HP:0002999
Congenital heart defects	HP:0001627
Renal alterations	HP:0000077

Occasional features

Seizures (HP:0001250), global developmental delay (HP:0001263), basal ganglia lesions (HP:0002134), aplasia cutis (HP:0001057), short thorax/sternum (HP:0005257), widely spaced/laterally displaced nipples (HP:0006610), cryptorchidism (HP:0000028), clubfoot (HP:0001762), hypotonia (HP:0001252), agenesis of the septum pellucidum (HP:0001331), and 13 pairs of ribs (HP:0000891). The first affected male exhibited "*agenesis of the septum pellucidum, a short chest and sternum, 13 pairs of ribs, a single hemivertebra, laterally displaced nipples, hydronephrosis, unilateral cryptorchidism, unilateral single palmar crease, bilateral clubfoot, and hypotonia*" ([PMID: 32869452](#)).

Severity/progression: Skeletal and craniofacial features are severe and static (congenital, non-progressive structural malformations). A milder facial phenotype has been documented in two sibs with compound heterozygous variants ([PMID: 35779070](#)), indicating variable expressivity. Neurologic features (seizures, basal-ganglia lesions) may be progressive and are an adverse prognostic sign.

Quality-of-life impact: Lifelong disability from short stature, joint dislocations, and skeletal dysplasia (mobility, orthopedic burden); hearing impairment from microtia; potential cardiac and renal morbidity; and neurodevelopmental disability in the CNS-affected subset. Formal QoL instruments (EQ-5D, SF-36) have not been applied given the disease's rarity.

4. Genetic / Molecular Information

Causal gene: *HSPA9* (Heat Shock Protein Family A member 9; HGNC:5244; OMIM 600548), located at chromosome 5q31.2, encoding *mortalin** (also mtHSP70, GRP75, PBP74). Disease OMIM #616854.

Pathogenic variant spectrum: Approximately 13 **pathogenic variants** have been catalogued ([PMID: 38284453](#)). All are **germline and biallelic** (homozygous in consanguineous/founder families, compound heterozygous otherwise), and classified pathogenic/likely pathogenic under ACMG/AMP criteria. Individually the alleles are rare or absent in gnomAD (carrier-level frequencies).

Variant (cDNA / protein)	Class	Domain	Reference
R126W	Missense	Nucleotide-binding domain (NBD)	PMID: 30933555
Y128C	Missense	Nucleotide-binding domain (NBD)	PMID: 30933555
c.955C>T (p.L319F)	Missense	—	PMID: 32869452
c.818T>G (p.L273X)	Nonsense (NMD)	—	PMID: 32869452
c.882_883delAG	Frameshift	— (recurrent, East Asian)	PMID: 36052765
c.613A>G	Missense	—	PMID: 36052765
c.1822-1G>A	Splice (pathogenic)	Substrate-binding domain (SBD)	PMID: 38281662
c.1411-3T>G	Splice (likely pathogenic)	Substrate-binding domain (SBD)	PMID: 38281662

"novel variants *c.818 T > G (p.L273X)* and *c.955C > T (p.L319F)* in the *HSPA9* gene" ([PMID: 32869452](#)); "*c. 1822-1G>A* and *c. 1411-3T>G* were classified as pathogenic and likely pathogenic, respectively" ([PMID: 38281662](#)).

Functional consequences: The overall mechanism is **loss of function / hypomorphism**. Truncating alleles undergo nonsense-mediated decay — "*qPCR analysis provides supporting evidence for a nonsense-mediated decay mechanism for the HSPA9 truncating variant*" ([PMID: 32869452](#)). Missense NBD alleles impair enzymatic/chaperone function: "*the surface mutations R126W and Y128C have far-reaching effects that disrupt ATP hydrolysis, interdomain linker binding, and thermostability and increase the propensity for aggregation*" ([PMID: 30933555](#)).

Modifier genes / epigenetics / chromosomal abnormalities: No disease modifiers, epigenetic mechanisms, or large-scale chromosomal abnormalities are described for EVEN-PLUS. (Note: heterozygous *HSPA9* deletion at 5q31.2 is separately implicated in del(5q) myelodysplastic syndrome — a distinct, somatic, haploinsufficiency context, [PMID: 21123823](#).)

5. Environmental Information

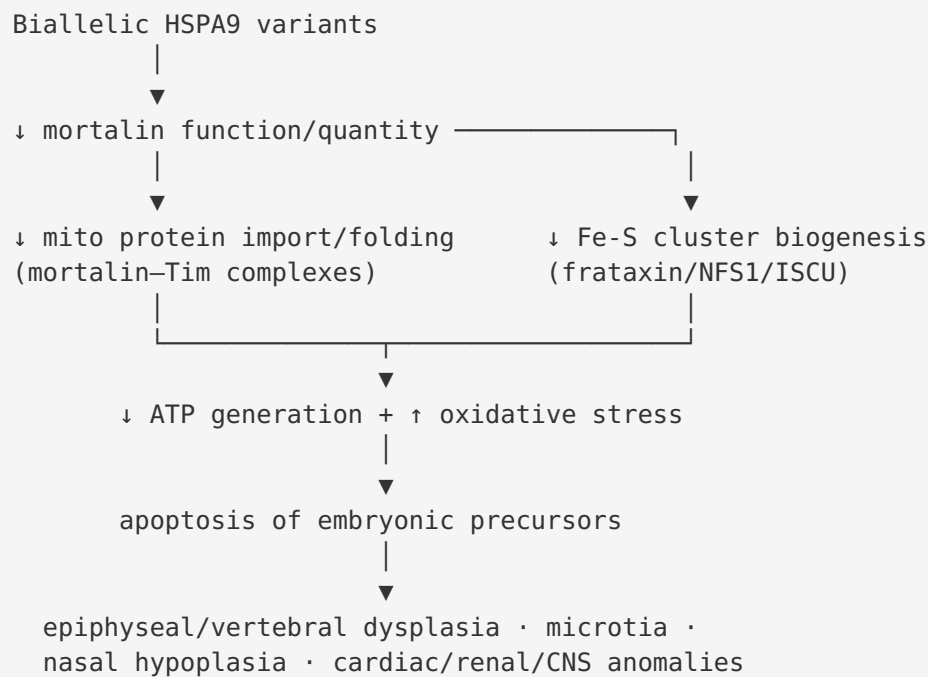
Not applicable. EVEN-PLUS is a purely Mendelian disorder. No environmental factors, toxins, radiation, occupational exposures, lifestyle factors, or infectious agents are known to cause, trigger, or modify the disease. The only non-genetic reproductive variable of note is consanguinity, which increases the likelihood of homozygosity for a founder allele.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic/LoF *HSPA9* variants** (missense, nonsense→NMD, frameshift, splice) *lead to* reduced quantity and/or impaired function of mortalin (mtHSP70/GRP75).
2. Reduced/dysfunctional mortalin *disrupts* ATP hydrolysis, interdomain (NBD↔SBD) communication, and thermostability, and *increases* the protein's aggregation propensity (demonstrated biophysically for R126W, Y128C — [PMID: 30933555](#)).
3. Impaired mortalin chaperone activity *results in* inefficient import and folding of nuclear-encoded mitochondrial proteins via mortalin–Tim complexes ([PMID: 17460192](#)) **and** (branch) *impairs* iron–sulfur (Fe-S) cluster biogenesis through mortalin's role as the Ssq1 homolog interacting with frataxin/ISD11/NFS1/ISCU ([PMID: 17331979](#)).
4. These deficits *lead to* inefficient mitochondrial biogenesis and energy (ATP) generation ([PMID: 17460192](#)), and *lead to* accumulation of oxidative stress (mortalin is a major oxidation target and oxidative-stress buffer).
5. Oxidative stress and energetic/proteostatic failure *result in* apoptosis of metabolically demanding, rapidly dividing embryonic precursor cells (inferred from the zebrafish model, where the analogous lesion "*compromises mitochondrial function, producing oxidative stress and apoptosis distinctly in blood cells*" — [PMID: 15650063](#)).

6. Precursor cell dysfunction/death during organogenesis *manifests as* the malformation phenotype — epiphyseal/vertebral skeletal dysplasia, microtia, nasal hypoplasia, and cardiac/renal/CNS anomalies. **(This final step from cellular defect to specific tissue malformation is inferred, not directly demonstrated in human tissue.)**
7. Complete loss *results in* embryonic lethality (mouse homozygous KO — [PMID: 25550197](#)); surviving human patients therefore retain partial (hypomorphic) mortalin function, defining a lethality gradient at the severe end of the allelic spectrum.



Mechanistic detail

- **Molecular pathways:** Mitochondrial protein import (TIM23/PAM machinery, mortalin as the matrix motor ATPase); Fe-S cluster (ISC) assembly pathway; HSP70 chaperone ATPase cycle. Mortalin also regulates the p53 pathway ([PMID: 30933555](#), [PMID: 25645922](#)). Suggested GO terms: **GO:0030150** (protein import into mitochondrial matrix), **GO:0016226** (iron-sulfur cluster assembly), **GO:0006457** (protein folding), **GO:0006979** (response to oxidative stress).
- **Cellular processes:** Apoptosis (GO:0006915), oxidative-stress response, and mitochondrial biogenesis failure. Cellular senescence has been linked to mortalin dysfunction ([PMID: 17460192](#)).

- **Protein dysfunction:** Loss of function via reduced expression (NMD) or catalytic/folding impairment; NBD mutants additionally show gain of **aggregation propensity** — a partial destabilization/misfolding phenotype (PMID: 30933555).
- **Metabolic changes:** Impaired oxidative phosphorylation / mitochondrial energy generation (inferred).
- **Immune involvement:** None as a primary mechanism.
- **Tissue damage mechanism:** Oxidative stress → apoptosis (demonstrated in the hematopoietic lineage in zebrafish; inferred for skeletal/craniofacial precursors in humans).

Cell types (suggested CL terms): chondrocyte (CL:0000138) and osteoblast (CL:0000062) for skeletal dysplasia; neural crest-derived cells for craniofacial (ear/nose) structures; cardiomyocyte (CL:0000746) and renal epithelial cells for organ malformations. These cell-type assignments are inferred from the affected anatomy rather than directly demonstrated.

7. Anatomical Structures Affected

Organ / system level (primary): - **Skeletal system** (UBERON:0001434): epiphyses (femoral head, knee), vertebrae (coronal clefts, hemivertebrae), acetabula, patellae, thorax/sternum, ribs. - **External/middle ear** (UBERON:0001690): microtia / dysplastic ears. - **Nose / nasal bone** (UBERON:0001705, UBERON:0002517): nasal hypoplasia, triangular nares. - **Face / eyebrow region** (UBERON:0001711): synophrys, arched eyebrows.

Secondary organ involvement: - **Cardiovascular system** (UBERON:0004535): congenital heart defects (HP:0001627). - **Renal/urinary system** (kidney UBERON:0002113): renal alterations, hydronephrosis (HP:0000077). - **Central nervous system / brain** (UBERON:0000955): basal-ganglia lesions (HP:0002134), agenesis of septum pellucidum (HP:0001331).

Tissue/cell level: Connective/skeletal tissue (cartilage, bone), with epiphyseal cartilage and growth-plate chondrocytes prominently affected; neural-crest-derived craniofacial mesenchyme (ear, nose).

Subcellular level: The primary compartment is the **mitochondrion** (GO:0005739), specifically the **mitochondrial matrix** (GO:0005759), where mortalin operates. Suggested GO cellular-component terms: GO:0005739 (mitochondrion), GO:0005759 (mitochondrial matrix), GO:0005758 (mitochondrial intermembrane space, for import).

Localization / lateralization: Skeletal features are generally **bilateral** and largely symmetric; some findings (cryptorchidism, single palmar crease, clubfoot) are reported unilaterally in individual patients (PMID: 32869452).

8. Temporal Development

Onset: Congenital / **prenatal**. *"It has a prenatal onset due to defects in the HSPA9 gene"* (PMID: 36052765). Malformations are established during embryonic/fetal organogenesis and are often detectable on prenatal ultrasound, prompting prenatal genetic diagnosis (PMID: 38281662).

Progression: The core skeletal and craniofacial malformations are **static/non-progressive** (structural, congenital). The disorder is chronic and lifelong for survivors. A lethality gradient exists at the severe end: some pregnancies are ascertained prenatally and terminated, and complete mortalin loss is embryonic-lethal in mice (PMID: 25550197).

Neurologic course: In the CNS-affected subset, seizures and basal-ganglia lesions may be progressive and represent an evolving morbidity beyond the static skeleton (PMID: 38284453).

Critical period: Embryonic organogenesis is the window of vulnerability. There is no post-natal therapeutic window to reverse established malformations; the only "intervention window" is **pre-conception/pre-implantation** (PGT) or prenatal.

9. Inheritance and Population

Epidemiology: Ultra-rare. Fewer than ~15 patients have been reported worldwide. Liu et al. collated *"12 cases (9 females)... from 6 relevant research items for analysis"* (PMID: 38284453). No formal prevalence or incidence estimate exists; Orphanet classifies it among ultra-rare bone dysplasias. The apparent female predominance (9/12) may reflect ascertainment/reporting bias in this tiny sample rather than a true sex bias (the disorder is autosomal).

Inheritance: **Autosomal recessive**, biallelic — homozygous in consanguineous/founder families, compound heterozygous otherwise.

Penetrance / expressivity: Complete penetrance for biallelic pathogenic genotypes; **variable expressivity** documented (milder facial phenotype in two sibs, PMID: 35779070).

Founder effects / population: A recurrent frameshift allele, **c.882_883delAG**, "*may have a higher distribution frequency in East Asian populations*" (PMID: 36052765), suggesting a founder-type allele. Consanguinity contributes to homozygosity in some families.

Carrier frequency: Not formally established; individual pathogenic alleles are rare or absent in gnomAD.

Anticipation / mosaicism: No genetic anticipation (not a repeat-expansion disorder). No germline mosaicism specifically reported.

10. Diagnostics

Genetic testing (definitive): Diagnosis is established by **whole-exome sequencing (WES)** or whole-genome sequencing identifying **biallelic HSPA9 variants**, confirmed by Sanger sequencing. "*HSPA9 compound heterozygous variants c.882_c.883delAG and c.613A>G were identified by exome sequencing*" (PMID: 36052765). WES/WGS is the highest-yield approach because the phenotype overlaps other skeletal dysplasias/mitochondrial chaperonopathies and single-gene testing may not be prioritized without genetic guidance. Targeted *HSPA9* testing or skeletal-dysplasia gene panels are appropriate confirmatory routes once the diagnosis is suspected.

Imaging (supportive): Characteristic radiographic/MRI findings — dysplastic/absent femoral-head epiphyses, "fork-shaped" (bifid) distal femur, dysplastic knee epiphyses and acetabula, vertebral coronal clefting, delayed ossification, hypoplastic nasal bone (PMID: 36052765, PMID: 32869452). Cerebral MRI may show basal-ganglia lesions and septum-pellucidum agenesis in CNS-affected patients.

Clinical criteria: No formal consensus diagnostic criteria exist. Diagnosis is gestalt (EVEN core features) plus molecular confirmation.

Differential diagnosis: The principal differential is **CODAS syndrome** (*LONP1*; MIM 600373), which shares epiphyseal, vertebral, and ocular changes but is distinguished from EVEN-PLUS by the latter's severe microtia and nasal hypoplasia — "*we delineate a similar but distinct condition that shares the epiphyseal, vertebral and ocular changes of CODAS but also included severe microtia, nasal hypoplasia*" (PMID: 26598328). Both are grouped as "mitochondrial chaperonopathies," alongside AIFM1-related spondyloepimetaphyseal dysplasia with

neurodegeneration — *"EVEN-PLUS syndrome caused by mutations of HSPA9 and CODAS syndrome due to LONP1 mutations"* (PMID: [27102849](#)). Other spondyloepiphyseal/spondyloepimetaphyseal dysplasias should also be considered.

Screening: No newborn or population screening exists (ultra-rare). Cascade carrier testing of relatives and reproductive-partner testing are appropriate once a familial variant is known.

11. Outcome / Prognosis

Survival/mortality: No formal survival statistics exist. There is a **lethality gradient**: the most severe genotypes are prenatally lethal or lead to pregnancy termination after prenatal diagnosis, and complete mortalin loss is embryonic-lethal in mice (PMID: [25550197](#), PMID: [38281662](#)). Postnatal survivors have a chronic, lifelong course.

Morbidity/function: Survivors carry a substantial, lifelong disability burden — skeletal dysplasia (short stature, joint dislocations, mobility limitation), hearing impairment (microtia), and potential cardiac and renal complications. Neurodevelopmental disability occurs in the CNS-affected subset.

Disease course / prognostic factors: CNS involvement is an **adverse prognostic feature**: *"Two patients had seizures and basal ganglia lesions in cerebral MRI"* (PMID: [38284453](#)). Genotype severity (residual mortalin function) is the principal determinant of phenotypic severity, ranging from prenatal lethality to milder facial-predominant presentations (PMID: [35779070](#)).

Recovery potential: None for the structural malformations — they are congenital and fixed. Management is supportive.

12. Treatment

There is no disease-modifying or curative therapy for EVEN-PLUS syndrome. Management is **supportive and multidisciplinary**, targeting the affected systems:

Domain	Supportive intervention	Suggested NCIT
Skeletal/orthopedic	Orthopedic management of dislocations, scoliosis, epiphyseal dysplasia; physical/occupational therapy	NCIT:C15329 (Orthopedic Procedure); NCIT:C15690 (Physical Therapy)
Audiologic	Hearing assessment and aids for microtia-associated hearing loss	NCIT:C15190 (Supportive Care)
Cardiac	Evaluation and management of congenital heart defects	NCIT:C15329
Renal	Monitoring/management of renal anomalies (e.g., hydronephrosis)	NCIT:C15190
Neurologic	Antiepileptic management; developmental support	NCIT:C15190

- **Pharmacotherapy / pharmacogenomics:** None disease-specific. Symptomatic only (e.g., antiepileptics for seizures).
- **Advanced therapeutics (gene, cell, RNA, targeted, immuno):** None available or in trials. Gene replacement/editing of *HSPA9* is a conceptual future avenue only (see Follow-up).
- **Surgical/interventional:** Orthopedic and cardiac surgery as indicated by structural anomalies.
- **Experimental treatments (NCT trials):** None identified for EVEN-PLUS.

The strongest "intervention" reported is **reproductive prevention** (see Section 13), not treatment of affected individuals.

13. Prevention

Because the malformations are congenital and untreatable, prevention is **reproductive/genetic** rather than clinical.

- **Genetic counseling:** For couples with a prior affected child or known carrier status — 25% recurrence risk per pregnancy (autosomal recessive). Cascade carrier testing of relatives is appropriate.

- **Prenatal diagnosis:** Detection of biallelic *HSPA9* variants (and/or ultrasound malformations) in at-risk pregnancies.
- **Preimplantation genetic testing (PGT):** The most effective demonstrated prevention. Chang et al. performed prenatal-to-preimplantation genetic diagnosis and selected mutation-free embryos: "*Assisted reproduction with mutation-free embryos successfully blocked the transmission of mutations*" (PMID: 38281662), resulting in a healthy birth.
- **Primary/secondary/tertiary prevention, immunization, public health, prophylaxis:** Not applicable (no environmental or infectious etiology). Tertiary prevention is limited to supportive management of complications.

14. Other Species / Natural Disease

HSPA9/mortalin is **deeply evolutionarily conserved**, and orthologs underpin the disease's model organisms, but no naturally occurring EVEN-PLUS-equivalent disease has been described in companion animals or wildlife.

Species	NCBI Taxon	Ortholog	Database
Human	9606	HSPA9	OMIM/HGNC
Mouse	10090	Hspa9	MGI
Zebrafish	7955	hspa9 / hspa9b	ZFIN
Yeast (<i>S. cerevisiae</i>)	4932	SSQ1 / SSC1	SGD

Mortalin/GRP75 is "*a homolog of the yeast ssq1 chaperone that integrates iron-sulfur clusters into imported mitochondrial proteins*" (PMID: 17331979), establishing conservation of the core Fe-S biogenesis and import functions from yeast to human. No zoonotic potential or cross-species transmission applies (non-infectious, genetic).

15. Model Organisms

Although no model was engineered specifically to phenocopy the EVEN-PLUS skeletal syndrome, existing *Hspa9* models establish the gene's essentiality and core mitochondrial mechanism.

Zebrafish — "crimsonless" (*hspa9b* mutant): A glycine-to-glutamate substitution in the substrate-binding domain recapitulates **ineffective hematopoiesis**: *"This mutation compromises mitochondrial function, producing oxidative stress and apoptosis distinctly in blood cells. Thus, we identify an essential role for Hspa9b in hematopoiesis and implicate both loss of HSPA9B specifically and mitochondrial dysfunction generally in the pathogenesis of the MDS"* (PMID: 15650063). This is the best mechanistic model linking mortalin loss to the mitochondrial-dysfunction → oxidative-stress → apoptosis cascade central to EVEN-PLUS pathophysiology, though it models a hematopoietic (MDS-relevant) rather than skeletal readout.

Mouse — *Hspa9* knockout: *"homozygous knockout of Hspa9 is embryonically lethal, mice with heterozygous deletion of Hspa9 (*Hspa9*(+/-)) are viable"* (PMID: 25550197). This establishes that complete mortalin loss is incompatible with life — implying EVEN-PLUS alleles are hypomorphic — and heterozygous models inform del(5q) MDS biology. Knockdown of *Hspa9* in mice reduces hematopoietic progenitors (PMID: 21123823).

Yeast — *SSQ1/SSC1*: The mortalin ortholog provides the biochemical foundation for Fe-S cluster integration into mitochondrial proteins (PMID: 17331979).

Model types available: knockout (mouse), point-mutant (zebrafish), and recombinant-protein/in-vitro biophysical systems (used to characterize R126W and Y128C — PMID: 30933555).

Model limitations: No existing model reproduces the full EVEN-PLUS skeletal/craniofacial malformation phenotype; current models capture hematopoietic and embryonic-lethality readouts rather than the epiphyseal-vertebral-ear-nose skeletal dysplasia. A conditional or knock-in model carrying EVEN-PLUS-specific hypomorphic alleles (e.g., R126W) would be needed to study skeletal pathogenesis directly.

Resources: MGI (mouse), ZFIN (zebrafish), SGD (yeast).

Mechanistic Model / Interpretation

EVEN-PLUS syndrome is best understood as a **developmental mitochondrial chaperonopathy**. The unifying molecular lesion is partial loss of mortalin, the mitochondrial matrix HSP70 that powers protein import and folding and supports Fe-S cluster assembly. Two independent lines of evidence converge on this model: (1) biophysical characterization of patient missense mutations (R126W, Y128C) showing loss of ATPase/chaperone competence and gain of aggregation propensity ([PMID: 30933555](#)), and (2) the essentiality of mortalin across species, where complete loss is embryonic-lethal (mouse) or produces mitochondrial-dysfunction-driven oxidative stress and apoptosis (zebrafish) ([PMID: 25550197](#), [PMID: 15650063](#)).

The critical inference is a **dose–severity continuum**: because null genotypes are lethal, viable EVEN-PLUS patients must retain some residual mortalin activity. The amount of residual function plausibly explains the observed phenotypic range — from prenatal lethality/termination at one extreme to the milder facial-predominant sib phenotype at the other ([PMID: 35779070](#)). The tissue selectivity (skeleton, ear, nose, heart, kidney, brain) is not yet mechanistically explained but is consistent with high mitochondrial/proteostatic demand in rapidly proliferating embryonic precursors (chondrocytes, neural-crest mesenchyme) during a narrow organogenesis window. The parallel to CODAS (LONP1, a mitochondrial protease) reinforces that disrupting *either* mitochondrial protein folding (mortalin) *or* mitochondrial protein turnover (LONP1) yields an overlapping epiphyseal-vertebral-craniofacial phenotype — implicating mitochondrial proteostasis broadly, rather than any single client protein, as the developmental bottleneck.

Evidence Base

PMID	Title (abbrev.)	Role in this report	Evidence type
26598328	<i>HSPA9 mutations cause EVEN-PLUS</i>	Delineates & names the disease; biallelic HSPA9; CODAS overlap; mitochondrial chaperonopathy	Human clinical + genetic
30933555	<i>Biophysical consequences of EVEN-PLUS mutations for mortalin</i>	Molecular mechanism of R126W/Y128C (ATP hydrolysis, aggregation)	In vitro / biophysical
38284453	<i>New phenotype in a Chinese family + review</i>	12-case phenotype spectrum & frequencies; CNS morbidity	Human clinical (case series/ review)
32869452	<i>Novel variants + HSPA9 dysfunction</i>	First affected male; NMD evidence; expanded HPO	Human clinical + in vitro
36052765	<i>Exome + clinical feature analysis</i>	Prenatal onset; East-Asian recurrent allele; WES diagnosis	Human clinical + genetic
38281662	<i>Prenatal to PGT diagnosis</i>	Splice variants; PGT prevention	Human clinical / reproductive
35779070	<i>Broadening phenotypic spectrum</i>	Milder phenotype; original EVE dysplasia family confirmed	Human clinical
27102849	<i>AIFM1 SEMD with neurodegeneration</i>	Groups EVEN-PLUS/CODAS as mitochondrial chaperonopathies	Human clinical (differential)
15650063	<i>Loss of Hspa9b in zebrafish</i>	Model: mito dysfunction → oxidative stress → apoptosis	Model organism
25550197	<i>Reduced Hspa9 in mouse B cells</i>	KO embryonic lethality; hypomorphism inference	Model organism
21123823	<i>Hspa9 del(5q31.2) knockdown</i>	Haploinsufficiency & hematopoiesis (distinct MDS context)	Model organism
17331979	<i>Frataxin-ISD11-chaperone interactions</i>	Mortalin/Ssq1 role in Fe-S biogenesis	In vitro / biochemical
17460192	<i>Mortalin in cellular senescence</i>	Import/chaperone/oxidative-stress functions	Review / in vitro

PMID	Title (abbrev.)	Role in this report	Evidence type
25645922	<i>Mortalin point mutations & PD</i>	p53 regulation, oxidative-stress tolerance	In vitro

Limitations and Knowledge Gaps

1. **Extreme rarity (n < 15).** All phenotype frequencies, sex ratios, and prognostic statements derive from small case series; the apparent female predominance (9/12) is likely an ascertainment artifact and should not be over-interpreted.
2. **The final mechanistic step is inferred.** The link from mortalin deficiency → mitochondrial/oxidative/apoptotic failure → *specific* skeletal and craniofacial malformations has not been demonstrated in human patient tissue or in a skeletal disease model. The causal chain from cellular defect to organ-specific malformation remains a plausible inference from the zebrafish hematopoietic model and general mortalin biology.
3. **No skeletal disease model.** Existing models (zebrafish hematopoiesis, mouse embryonic lethality) do not recapitulate the EVEN core features, limiting mechanistic and therapeutic study.
4. **Genotype–phenotype correlations are preliminary.** With only ~13 variants reported, the relationship between residual mortalin activity and phenotypic severity (including the lethality gradient) is inferred rather than quantitatively established.
5. **No natural-history data, QoL instruments, or formal survival statistics** exist.
6. **No epidemiologic prevalence/incidence estimate** and no carrier-frequency data beyond gnomAD-level rarity.

Proposed Follow-up Experiments / Actions

1. **Generate an allele-specific mouse or zebrafish knock-in** carrying an EVEN-PLUS hypomorphic variant (e.g., R126W) to test whether skeletal/craniofacial malformations are recapitulated, and to define the affected cell types (chondrocytes, neural-crest mesenchyme) directly.

2. **Patient-derived iPSC → chondrocyte / neural-crest organoid models** to measure mitochondrial import efficiency, Fe-S cluster assembly, OXPHOS capacity, ROS, and apoptosis in the disease-relevant lineages, confirming the inferred causal chain in human cells.
3. **Systematic genotype–phenotype and residual-function correlation:** express the full reported variant panel in vitro (ATPase, thermostability, aggregation, import assays) and correlate residual activity with clinical severity to build a predictive severity model.
4. **Establish an international patient registry** to aggregate phenotype frequencies, natural history, survival, and QoL across the world's <15 (and growing) cases.
5. **Population screening for the recurrent East-Asian allele c.882_883delAG** to estimate carrier frequency and support targeted reproductive counseling.
6. **Explore proteostasis/antioxidant modulators** (e.g., small-molecule chaperone stabilizers, mitochondrial antioxidants) as conceptual therapeutic probes in the cellular/organoid models — recognizing that congenital structural malformations are unlikely to be reversible postnatally, so any therapeutic benefit would likely target progressive (e.g., neurologic) rather than static features.
7. **Refine ontology annotations** (submit the HPO frequencies and CL/UBERON/GO term mappings proposed here) to the disease knowledge base and to Orphanet/OMIM.

Consensus Answer

EVEN-PLUS syndrome (OMIM #616854) is an ultra-rare, autosomal-recessive, prenatal-onset malformation and skeletal-dysplasia syndrome caused by biallelic loss-of-function/hypomorphic variants in *HSPA9*, encoding the mitochondrial HSP70 chaperone mortalin (mtHSP70/GRP75). Deficient mortalin impairs mitochondrial protein import, Fe-S cluster biogenesis, and proteostasis — producing oxidative stress and apoptosis in embryonic precursors that manifest as epiphyseal/vertebral dysplasia, microtia, nasal hypoplasia, and cardiac, renal, and CNS anomalies — placing it among the "mitochondrial chaperonopathies" that overlap CODAS syndrome (*LONP1*). No disease-modifying therapy exists; care is supportive and prevention is reproductive (prenatal diagnosis or preimplantation genetic testing).